Marking Scheme

2026 · Paper Paper 1 · 105 marks

General Rules
- M / A / PROOF mark types
- **f.t.**： M marks always follow through. A marks follow through ONLY when isFollowThrough = true.
- Deduction cap： A(1)=1 · A(2)=1 · B=0 (u, pp)
- No deduction in any step scoring 0.
- A correct answer merits all marks unless a method is specified.
- Different notation is NOT penalized. · Benefit of doubt
- All fractional answers must be simplified.

Section A(1) (35 marks)

1. Laws of Indices [B-]

a (3 marks)

M1 $(x^3y^{-2})^{-2} = x^{-6}y^4$

M1 $(x^{-1}y)^3 = x^{-3}y^3; \dfrac{x^{-6}y^4}{x^{-3}y^3} = x^{-3}y$

A1 $\dfrac{y}{x^3}$

2. Linear Equations [B-]

a (3 marks)

M1 $x - y = 96 and 5y = x$

M1 $5y - y = 96 \Rightarrow 4y = 96$

A1 $y = 24$

3. Algebraic Fractions [B-]

a (3 marks)

M1 $2x - 6 = 2(x-3), so the L.C.M. of the denominators is 2(x-3)$

M1 $= \dfrac{4}{2(x-3)} + \dfrac{5}{2(x-3)} = \dfrac{9}{2(x-3)}$

A1 $\dfrac{9}{2x-6}$

4. Factorization [B]

a (2 marks)

M1 $4m^2 - 12m + 9 = (2m)^2 - 2(2m)(3) + 3^2$

A1 $= (2m-3)^2$

b (2 marks)

M1 $= (2m-3)^2 - (5n)^2$ f.t.

1M for using (a) and difference of two squares

A1 $= (2m-5n-3)(2m+5n-3)$ f.t.

5. Percentages [B]

a (4 marks)

M1 $Cost = \dfrac{160}{20} = \backslash 800$

M1 $Selling price = 800 + 160 = \backslash 960$

M1 $Let the marked price be \backslash M; M(1 - 25$

A1 $M = \backslash 1280$

u-1 if the dollar sign is missing

6. Compound Inequalities [B]

a (3 marks)

M1 $-3x+3 \geq x+19 \Rightarrow -4x \geq 16 \Rightarrow x \leq -4$

M1 $5x < -10 \Rightarrow x < -2$

A1 $Takingtheunion: x < -2$

b (1 mark)

A1 -3 f.t.

---

7. Transformation and Coordinates [B]

a (2 marks)

A1 $U' = (6, 8)$

A1 $V' = (-5, -9)$

b (2 marks)

M1 $\text{Slope} = \frac{-9-8}{-5-6}$ f.t.

A1 $= \frac{17}{11}$ f.t.

---

8. Congruent Triangles [B+]

a (2 marks)

M1 $AB = CD(given), \angle ABD = \angle CDB(given), BD = DB(commonside)$

PROOF $\therefore \triangle ABD \cong \triangle CDB(SAS)$

b (3 marks)

M1 $\angle CBD = \angle ADB = 27°(corr.$ \angles, $\cong$ \triangles$)$ f.t.

M1 $In\triangle BCD: \angle BDC = 180° - 94° - 27° = 59°(\angle sumof\triangle)$

A1 $\angle ADC = \angle ADB + \angle BDC = 27° + 59° = 86°$ f.t.

---

9. Frequency Distribution [B]

a (3 marks)

A1 $x = 22 - 6 = 16$

M1 $\text{Mean} = \frac{6(12)+16(17)+14(22)+4(27)}{40} = \frac{760}{40}$ f.t.

A1 $= 19hours$ f.t.

u-1 if unit missing

b (2 marks)

M1 $P = \frac{22}{40}$ f.t.

A1 $= \frac{11}{20}$ f.t.

Section A(2) (35 marks)

10. Partial Variation and Quadratic Equations [A-]

a (3 marks)

M1 $Letf(x) = hx^2 + mx, whereh \neq 0andm \neq 0$

M1 $9h + 3m = 45and4h - 2m = 10$

A1 $h = 4, m = 3 \Rightarrow f(x) = 4x^2 + 3x$

b (1 mark)

A1 $6f(x) = 6x(4x+3) = 0 \Rightarrow x = 0orx = -\frac{3}{4}$ f.t.

c (2 marks)

M1 $4x^2 + 3x - k = 0; \Delta = 3^2 - 4(4)(-k) = 9 + 16k > 0$ f.t.

A1 $k > -\frac{9}{16}$ f.t.

---

11. Stem-and-Leaf Diagram and Dispersion [A-]

a (3 marks)

M1 $Thereare16data; median = \frac{8\text{th} + 9\text{th}}{2} = \frac{29 + (30 + a)}{2} = 31$

A1 $a = 3$

A1 $Q_1 = \frac{20 + 23}{2} = 21.5, Q_3 = \frac{41 + (40 + b)}{2}; Q_3 - 21.5 = 21 \Rightarrow b = 4$

b (2 marks)

M1 $Themodeis23, whichoccurs3times; everyotherageoccursatmostonce$ f.t.

A1 $Afteroneplayerleaves, 23stilloccursatleasttwice, sothemoderemains23 : NOchange$

c (2 marks)

M1 $Range = 48 - 15 = 33; itdecreasesonlyifthe playeraged15ortheplayeraged48leaves$ f.t.

A1 $Removing15givesstandarddeviation \approx 10.8, removing48gives \approx 10.8; thegreatestis10.8(corr. to3sig. fig.)$ f.t.

r.t. 10.8

---

12. Circles and Coordinate Geometry [A-]

a (3 marks)

M1 $G = \left(\frac{80}{2}, \frac{60}{2}\right) = (40, 30)$

M1 $GH = \sqrt{(45 - 40)^2 + (42 - 30)^2} = \sqrt{25 + 144}$ f.t.

A1 $GH = 13$ f.t.

b (4 marks)

M1 $GP = radius = \sqrt{40^2 + 30^2 + 4556} = 84(aconstant)$

A1 $Area = \frac{1}{2}(GH)(GP)\sin\angle HGPisgreatestwhen\sin\angle HGP = 1, i.e. GH \perp GP$

M1 $HP = \sqrt{13^2 + 84^2} = \sqrt{7225} = 85$ f.t.

A1 $Perimeter = 13 + 84 + 85 = 182$ f.t.

---

13. Mensuration and Similarity [A-]

a (3 marks)

M1 $Ratioofvolumes = 27 : 64 \Rightarrow ratioofradii = 3 : 4$

M1 $Radiusofthesmallersphere = \frac{3}{4}(12) = 9 \text{ cm}$

A1 $Surfacearea = 4\pi(9)^2 = 324\pi \text{ cm}^2$ f.t.

b (4 marks)

M1 $Totalvolume = \frac{4}{3}\pi(9)^3 + \frac{4}{3}\pi(12)^3 = 972\pi + 2304\pi = 3276\pi \text{ cm}^3$

M1 $VolumeofA = \frac{1}{3}\pi(9)^2(12) = 324\pi \text{ cm}^3$

M1 $VolumeofB = 3276\pi - 324\pi = 2952\pi \text{ cm}^3$ f.t.

A1 $IfAandBweresimilar, theratioofradii = 18 : 9 = 2 : 1wouldgivevolumeofB = 8(324\pi) = 2592\pi \neq 2952\pi, sotheclaimisNOTcorrect$ f.t.

---

14. Polynomials [A]

a (3 marks)

M1 $p(1) = 3 \Rightarrow 3 + a + b - 24 = 3 \Rightarrow a + b = 24$

M1 $p(-2) = 0 \Rightarrow -24 + 4a - 2b - 24 = 0 \Rightarrow 2a - b = 24$

A1 $a = 16, b = 8$

b (2 marks)

M1 $p(2) = 3(8) + 16(4) + 8(2) - 24 = 24 + 64 + 16 - 24 = 80$ f.t.

A1 $p(2) \neq 0, sobythefactortheoremx - 2isNOTafactorofp(x)$ f.t.

c (3 marks)

M1 $p(x) = (x + 2)(3x^2 + 10x - 12)$ f.t.

M1 $For3x^2 + 10x - 12 = 0 : \Delta = 10^2 - 4(3)(-12) = 244 > 0and244isnotaperfectsquare$

A1 $Sothisquadratichastwoirrationalrootswhilex = -2isrational; theclaimisAGREED$ f.t.

Section B (35 marks)

15. Probability [A-]

a (2 marks)

M1 $P = \frac{C_2^7 \times C_2^9}{C_4^{16}} = \frac{21 \times 36}{1820}$

A1 $= \frac{27}{65}$

b (2 marks)

M1 $P = 1 - \frac{27}{65}$ f.t.

1M for using the complementary event / 使用互補事件

A1 $= \frac{38}{65}$ f.t.

---

16. Quadratic Functions [A-]

a (2 marks)

M1 $g(x) = 2(x^2 + 6kx) + 20k^2 + 6 = 2(x + 3k)^2 - 18k^2 + 20k^2 + 6$

A1 $g(x) = 2(x + 3k)^2 + 2k^2 + 6; vertex = (-3k, 2k^2 + 6)$

b (3 marks)

M1 $3g(-x) = 6x^2 - 36kx + 60k^2 + 18 = 6(x - 3k)^2 + 6k^2 + 18$ f.t.

A1 $B = (3k, 6k^2 + 18)$ f.t.

A1 $M = \left(\frac{-3k + 3k}{2}, \frac{(2k^2 + 6) + (6k^2 + 18)}{2}\right) = (0, 4k^2 + 12)$ f.t.

---

17. Roots of Equations and Sequences [A]

a (3 marks)

M1 $\alpha + \beta = -c and \alpha\beta = -12$

M1 $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

A1 $= c^2 + 24$

b (4 marks)

M1 $Common difference = (c^2 + 24) - c^2 = 24$ f.t.

A1 $3rd term = c^2 + 48 = 76 \Rightarrow c^2 = 28; the 1st term is 28$

M1 $S(n) = \frac{n}{2}[2(28) + (n - 1)(24)] = 12n^2 + 16n > 3 \times 10^5$ f.t.

A1 $S(499) = 2995996 < 3 \times 10^5 \cdot 10 and S(500) = 3008000 > 3 \times 10^5 \cdot 10; least n = 500$ f.t.

---

18. Trigonometry in 3-D [A]

a (3 marks)

M1 $By the cosine formula: BC^2 = 24^2 + 20^2 - 2(24)(20)\cos 105°$

M1 $BC^2 = 976 + 248.47 = 1224.47$

A1 $BC \approx 35.0$ cm

u-1 if unit cm missing

r.t. 35.0

b (4 marks)

M1 $Vertical height of C above the ground = AC \sin 60° = 20 \sin 60° \approx 17.32$ cm

M1 $B lies on the ground, so the height of N (mid-point of BC) = \frac{17.32}{2} \approx 8.66$ cm f.t.

M1 $The horizontal distance from A to N is found to be \approx 10.34$ cm f.t.

A1 $\tan\theta = \frac{8.66}{10.34} \Rightarrow \theta \approx 40.0° > 35°, so the claim IS correct$ f.t.

r.t. 40.0

---

19. Circles and Tangents [A+]

a (2 marks)

M1 $SlopeofAG = \frac{120-40}{90-30} = \frac{4}{3}$

A1 $y - 40 = \frac{4}{3}(x-30) \ \Rightarrow \ 4x - 3y = 0$

b (3 marks)

M1 $AG = \sqrt{60^2 + 80^2} = 100$

M1 $PQ \perp AG; letMbetheintersection. Inright-angled\triangle GPA, GM = \frac{GP^2}{GA} = \frac{50^2}{100} = 25$

A1 $M = (30,40) + 25\left(\frac{3}{5}, \frac{4}{5}\right) = (45,60)$ f.t.

c (4 marks)

M1 $AP = \sqrt{100^2 - 50^2} = 50\sqrt{3}andPQ = 2 \times \frac{GP \cdot AP}{GA} = 50\sqrt{3}$ f.t.

M1 $AP = AQ = PQ, so\triangle APQisequilateralanditsin-centreisthecentroid, lyingonAM$

M1 $AM = \sqrt{45^2 + 60^2} = 75; in-centre = A + \frac{2}{3}(M-A) = (60,80)andin-radius = \frac{75}{3} = 25$ f.t.

A1 $(x-60)^2 + (y-80)^2 = 625, i.e.\, x^2 + y^2 - 120x - 160y + 9375 = 0$ f.t.

d (3 marks)

M1 $Fortheequilateral\triangle APQthecentroid, in-centreandcircumcentrecoincideat(60,80)$ f.t.

M1 $Circumradius = \frac{2}{3}(AM) = 50, in-radius = 25$ f.t.

A1 $Ratioofareas = \pi(25)^2 : \pi(50)^2 = 1:4, sotheclaimisAGREED$ f.t.